

Commodity Return Forecasting

Predicting 1-Week and 1-Month Returns
Using Machine Learning

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1 Introduction

This report presents a systematic approach to forecasting short-term commodity returns using machine learning. The objective is to predict 1-week and 1-month returns from 60 anonymised features, with a strict methodology designed to prevent data leakage and produce realistic, out-of-sample performance estimates.

The key methodological principles are:

1. All preprocessing (imputation, scaling) is fitted on training data only.
2. Cross-validation uses temporal splits exclusively — no random K-fold.
3. Target lags are shifted by the forecast horizon to avoid overlap.
4. An embargo gap separates train and test sets in walk-forward folds.
5. All reported metrics are computed on non-overlapping observations.

The analysis progresses from simple linear baselines (ElasticNet) to gradient-boosted trees (XGBoost) with engineered features, demonstrating that careful feature engineering improves signal quality beyond what the model can extract from raw inputs alone.

2 Data Description

The dataset consists of daily observations with 60 anonymised features and two target variables: 1-week forward returns (`output1_1w`) and 1-month forward returns (`output1_1m`). Features are provided without labels, preventing economic interpretation but allowing a purely statistical approach to signal extraction.

Table 1: Dataset overview

Property	Value
Features	60 (anonymised)
Targets	1-week return, 1-month return
Missing values	~15% overall (varies by feature)
Horizons	5 trading days (1w), 21 trading days (1m)

3 Exploratory Data Analysis

3.1 Target Distributions

Both target distributions are approximately symmetric around zero with heavy tails (excess kurtosis), consistent with financial return distributions. The 1-month target exhibits higher variance, as expected for a longer horizon. Augmented Dickey-Fuller tests confirm stationarity of both series ($p < 0.01$).

3.2 Feature–Target Correlations

Individual feature–target correlations are generally weak ($|r| < 0.15$), which is expected for financial data: the signal is distributed across many features rather than concentrated in a few. This motivates the use of ensemble tree models that can combine many weak predictors.

3.3 Predictiveness Stability

Rolling correlation analysis (252-day window) reveals that feature predictiveness is not stable over time. Features that are strong predictors in one period may lose predictive power in another.

This non-stationarity motivates walk-forward evaluation rather than a single train/test split, and suggests that adaptive or regime-conditional models could further improve performance.

3.4 Feature Redundancy

The feature correlation matrix reveals clusters of highly correlated features ($|r| > 0.9$). This redundancy is partially addressed by XGBoost’s built-in feature selection and by L1 regularisation in ElasticNet. The engineered features (z-scores, ranks) provide alternative representations that reduce sensitivity to multicollinearity.

4 Feature Engineering

A total of 140+ features were engineered from the original 60, using only backward-looking transformations to prevent any form of look-ahead bias:

- **Rolling z-scores** (21d, 63d): Normalise each feature by its recent rolling mean and standard deviation, capturing whether current values are unusual relative to the recent regime.

$$z_t^{(w)} = \frac{x_t - \bar{x}_{t-w:t}}{\sigma_{t-w:t} + \epsilon} \quad (1)$$

- **Cross-sectional percentile ranks**: Rank each feature value relative to all other features on the same date, capturing relative positioning across the feature space. This is robust to outliers and non-stationarity in levels.
- **Rolling volatility**: Standard deviation over 21 days for the top 10 most correlated features, capturing changes in feature stability that may signal regime transitions.
- **Feature lags**: Lagged values of top features, shifted by the forecast horizon h to avoid overlap:

$$x_t^{\text{lag}_k} = x_{t-(h+k)}, \quad k \in \{0, 1, 2, 5, 10\} \quad (2)$$

- **Momentum features**: Change in feature value over the forecast horizon, measuring the recent trend in each predictor:

$$\Delta x_t^{(h)} = x_t - x_{t-h} \quad (3)$$

These transformations increase the feature space from 60 to approximately 200 features. XGBoost’s internal feature selection via split gain and L2 regularisation handle the increased dimensionality without explicit feature selection.

5 Modeling Approach

5.1 Model Progression

Three model classes are evaluated in a principled progression from simple to complex:

1. **ElasticNet**: Linear baseline with combined L1/L2 regularisation. Internal cross-validation uses `TimeSeriesSplit` with 3 folds. Preprocessing (mean imputation + standard scaling) is fitted on train data only via a scikit-learn `Pipeline`.
2. **XGBoost (raw features)**: Gradient-boosted trees on the original 60 features plus horizon-shifted target lags. XGBoost handles missing values natively, eliminating the need for imputation.
3. **XGBoost (engineered features)**: Same architecture on the expanded feature set (~ 200 features). This tests whether explicit feature engineering adds predictive power beyond what the tree model discovers through its own splits.

5.2 Evaluation Framework

All models are evaluated exclusively through 5-fold walk-forward cross-validation. Each fold uses a growing training window and a fixed-size test window, with an embargo gap of h days (where h is the forecast horizon) between the last training observation and the first test observation:

$$\text{Train}_k = \{1, \dots, T_k - h\}, \quad \text{Test}_k = \{T_k + 1, \dots, T_{k+1}\} \quad (4)$$

No single train/test split results are reported, as they are sensitive to the arbitrary choice of split point and can be misleadingly optimistic or pessimistic.

5.3 Metrics

Primary evaluation metrics are:

- **Pearson IC** (information coefficient): Correlation between predicted and realised returns. Computed on non-overlapping observations only (every h -th observation).
- **Hit rate**: Fraction of correct directional predictions (non-overlapping).
- **Annualised Sharpe ratio**: Mean PnL / Std PnL $\times \sqrt{252/h}$, computed on non-overlapping observations after transaction costs.

R-squared is deliberately not used as a primary metric, as it is less interpretable for signal-based trading strategies where even small positive correlations can be economically significant.

6 Walk-Forward Backtest

6.1 Position Sizing

Positions are sized proportionally to signal strength rather than using a binary sign function:

$$w_t = \text{clip} \left(\frac{\hat{y}_t}{\hat{\sigma}_{\hat{y}}}, -1, 1 \right) \quad (5)$$

where $\hat{y}_t$ is the predicted return, $\hat{\sigma}_{\hat{y}}$ is the standard deviation of predictions, and the clip function constrains positions to $[-1, 1]$. This ensures that stronger signals receive larger allocations while capping maximum exposure.

6.2 Transaction Costs

A cost of 2 basis points per unit of turnover is applied:

$$\text{Cost}_t = \frac{2}{10000} \times |w_t - w_{t-1}| \quad (6)$$

Net PnL is then $\text{PnL}_t^{\text{net}} = w_t \cdot r_t - \text{Cost}_t$, where r_t is the realised return.

6.3 Risk Metrics

Beyond Sharpe ratio, the following risk metrics are reported:

- **Maximum drawdown**: Largest peak-to-trough decline in cumulative PnL.
- **Calmar ratio**: Annualised return divided by maximum drawdown.
- **Win/loss ratio**: Average profit on winning trades divided by average loss on losing trades.
- **Average daily turnover**: Mean absolute change in position.

7 Results

7.1 Main Results

Table 2: Walk-forward results — XGBoost with engineered features (non-overlapping metrics)

Metric	1-Week	1-Month
Pearson IC	0.45	0.41
Hit Rate	66–67%	63%
Annualised Sharpe	2.2	1.6
Transaction Costs	2 bps	2 bps

7.2 Model Comparison

Table 3: IC progression across models (approximate, averaged across walk-forward folds)

Model	IC (1w)	IC (1m)	Improvement
ElasticNet	~0.30	~0.25	Baseline
XGBoost (raw features)	~0.38	~0.35	+25%
XGBoost (engineered)	~0.45	~0.41	+50%

Feature engineering contributes approximately 15–20% improvement in IC over raw features, demonstrating that domain-informed transformations add value beyond what the tree model discovers through its own splits. The largest improvements come from rolling z-scores and momentum features.

7.3 Interpretation

An information coefficient of 0.45 on the 1-week horizon is strong by the standards of financial return prediction, where ICs above 0.05–0.10 are generally considered exploitable at scale. The hit rate of 66% confirms directional predictability well above the 50% random baseline.

The annualised Sharpe ratio of 2.2 (1-week) after transaction costs indicates a high-quality signal. However, these results should be interpreted with caution: they apply to a single commodity with anonymised features, and real-world implementation would face additional challenges including market impact, liquidity constraints, and regime changes.

Performance varies across walk-forward folds, with some folds producing Sharpe ratios below 1.0. This heterogeneity is consistent with regime-dependent predictability and highlights the importance of reporting fold-level statistics rather than aggregate averages alone.

8 Feature Importance Analysis

SHAP (SHapley Additive exPlanations) values are computed on the last walk-forward fold to understand which features and engineered transformations drive predictions. Unlike split-based importance, SHAP provides both magnitude and direction of each feature’s contribution.

Key findings:

- Rolling z-scores and momentum features appear prominently among the top 20 predictors, confirming the value of feature engineering.

- Target lags contribute meaningfully but do not dominate the importance ranking, suggesting the signal is not purely autoregressive.
- Several raw features remain important even after engineering, indicating they carry information not fully captured by the transformations.
- Interactions between raw features and their z-score variants suggest the model benefits from seeing both level and regime-normalised representations.

9 Limitations and Future Work

9.1 Limitations

- **Single asset:** All results apply to one commodity. Cross-asset generalisation is untested; the signal may be commodity-specific.
- **Anonymised features:** The absence of feature labels prevents economic interpretation, making it impossible to assess whether the model captures known risk premia or novel patterns.
- **Regime dependence:** Performance varies meaningfully across walk-forward folds. Residual autocorrelation in some folds indicates unmodelled temporal structure.
- **No hyperparameter optimisation:** XGBoost parameters are set to reasonable defaults but not optimised per fold. Overfitting risk is mitigated by regularisation but not eliminated.
- **No market impact modelling:** The backtest assumes fills at the 2 bps cost level regardless of position size, which may understate real-world execution costs for larger allocations.

9.2 Future Work

1. **Hyperparameter tuning:** Bayesian optimisation (e.g. Optuna) on walk-forward folds, with the outer fold providing an unbiased estimate of generalisation.
2. **Ensemble methods:** Blend XGBoost and ElasticNet predictions, potentially weighting by inverse fold-level variance.
3. **Regime detection:** Use volatility clustering or hidden Markov models to build conditional models that adapt to market regimes.
4. **Multi-asset extension:** Test whether the signal and feature engineering pipeline generalise across commodities.
5. **Alternative architectures:** LightGBM, CatBoost, and temporal convolutional networks for capturing longer-range dependencies.

10 Conclusion

This project demonstrates that commodity returns at the 1-week and 1-month horizons contain exploitable predictive signal. A walk-forward validated XGBoost model with engineered features achieves an information coefficient of 0.45 (1-week) and 0.41 (1-month), with hit rates above 63% and annualised Sharpe ratios of 1.6 to 2.2 after transaction costs.

The central methodological contribution is the anti-leakage discipline applied throughout. Without temporal splits, horizon-shifted lags, embargo gaps, and non-overlapping metrics, the same data can produce dramatically inflated performance numbers. Every design choice in this pipeline was made to ensure that reported results would be reproducible in a live trading environment.

Feature engineering contributes meaningfully: rolling z-scores, cross-sectional ranks, and momentum features improve IC by 15–20% over raw inputs. The progression from linear to tree-based models shows diminishing but real returns to model complexity. The signal is real but regime-dependent, and any production deployment would require continuous monitoring and periodic

recalibration.